

piezobuzzer (PZ1) at the output of IC1 starts sounding for a fixed duration, determined by the values of resistor R7 and capacitor C5. PZ1 can be replaced with an electromagnetic relay to drive heavy external electrical loads such as power sirens.

Assemble the circuit on a general-purpose PCB and enclose it (including battery) in a tamper-proof cabinet. Glue the condenser microphone at the rear side of the window/door glass to be protected, and connect the microphone to the sensor circuit using

a short length of transparent screened cable.

EFY note. Using a glass break sensor may cause false alarms by confusing the breaking of glass such as cookware, or the sound of bells, with the sound of breaking windows. ●

MILLIOHM METER

■ RAJU BADDI

Milliohm meter is used to check low resistance of connecting wires. It can deduce the resistance of thick wires, condition of contacts and the length or quality of wires. Although the continuity can be easily checked using a continuity tester, the exact condition of the circuit cannot be assessed without proper measurement. For example, a switch in good condition will have very low contact resistance, whereas, if the switch is in bad condition its resistance would be much higher even though it may pass the continuity test.

The quality of wires can also be assessed. Broken strands can significantly increase the resistance on the milliohm scale. Commercial copper wire has resistivity (ρ) as follows:

$$\rho = 17.2 \times 10^{-9} \text{ ohm metre, at } 20^\circ\text{C}$$

So, a 0.5mm diameter wire having a length of 10 cm would have a resist-

ance of ~9 milliohm. This is calculated using the following standard formula:

$$\begin{aligned} \text{Resistance (R)} &= \text{Resistivity} \times \frac{\text{Length of wire}}{\text{Area of cross-section of the wire}} \\ &= \rho \times \frac{l}{a} \end{aligned}$$

You can use this relation to calculate resistance of different metals. The resistance of ordinary connecting copper wires in electronic circuits is of the order of milliohms. To measure it, a milliohm meter is required. The knowledge of the resistance of a certain wire of unknown length can be used to deduce its length or vice-versa. Knowing the length of the wire in certain winding of a transformer or a rotor, and assuming certain standard winding configuration, one can deduce the approximate number of turns in that winding. All these require a meas-

urement of the resistance of the wire which is of the order of milliohms and above.

The circuit described here has a minimum resolution of 50 milliohms and a maximum range of 10 ohms. These limits seem to be adequate to deal with ordinary conducting wires. The operation of the circuit is based on the standard technique; pass a known fixed current through the resistance (R_x) and measure the potential drop across it.

Since the resistances being measured are extremely small compared to the other electronic standards of hundreds and thousands of ohms, a fixed-precision resistance is used in series with the supply voltage to get a constant current. A 50mA constant current is obtained by using a resistance of 100 ohms and a supply of 5V. This will be used to measure a maximum resistance of 10 milliohm which is 10000^{th} of 100 ohms and so its presence in the circuit is negligible compared to the 10 ohms resistance. Thus the current remains sufficiently constant throughout any range.

The voltage-drop across the unknown resistance (R_x) is measured using op-amp TL081, which has been configured as a high-gain non-inverting DC amplifier with a gain of $100 = (R_f/R_i + 1)$. The voltage drop across R_x is multiplied by 100 and presented to the meter by the op-amp.

The meter's resistance has been assumed to be 700 ohms, so an additional 500 ohms is required to properly calibrate the instrument. This can be done using preset VR2 in series with the

